SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I L.Sathyam Sharma (192526O282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L.Sathyam

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

L. sathyam sharma
CLOUD COMPUTIND
CSA01519
192526028 2
CO2 - AT2

1 Auto scaling and load Balancing:-

Auto scaling automatically increases or decreases cloud resources according to users demands. load balancing distributes incoming traffic across multiple servers to prevent any single from becoming overloaded These technologies reduce downtime, improve responds speeds, and provide user experience monitoring tools tracks cpu usage, traffic and server performance to identify problems and trigger scaling

Example:-

During an Amazon Great India Festival sale, auto scaling adds servers when traffic increases, while load balancing distributes customer request across them. This helps prevent website crashes and delays

2 Secure cloud Storage and Backup:-

cloud storage protects backup data through redundancy, storing multiple copies of data across different servers or location. This improves durability and allows data recovery if one server fails

Encryption protects data from unauthorized access, while access controll allows only authorized access, while access controll allows only authorized users to view or modify views cloud storage also

Pay for the storage they use instead of purchasing expensive hardware

Example:- If a company's local server is damaged by a fire, it can restore its files and databases from cloud backups stored in another region ensuring buisness continuity

3 CDN and Global Application Performance:-

A content delivery Network is a group of distributed servers that delivers contents from locations closers to users. Edge servers store cached content near users, reducing latency and improving loading spins speeds cloud providers supports global availability through multiple regions, data replication, and break up servers This helps applications remains accessible even if one location experiences problem

Example:-
Netflix uses content delivery technology to deliver videos from nearby servers. This reduces buffering and provides smoother streaming for users in different countries

4 cloud security and Regulatory compliance:-

cloud providers supports regulatory compliance through security standards audit reports, access controls, and compliance tools security standards audit reports and logging and record user activities logging and detect suspicious transactions while encrypting protects sensitivie data during storage and transmisson

Data Sovereignity can be maintained by selecting cloud regions that meet the country's data storage and Process requirements

example:-
A bank can encrypt customer account details, restricts access to authorized employees and record every online transaction. These measure help protect financial information, detect fraud and support regulatory audits.

5.) containers and virtual machines:-

containers package an application with its required libaries and dependencies. They share the host operating systems, requiring more memory and storage but providing stronger isolation in many configurations

Orchestration tools such as kubernetes automates deployment scaling, load distribution and recovery of failed containers. containers are also portable across different cloud environments

Examples:-

An online shopping application can use separate containers for login, payments, product search, and order management. Kubernetes can scale the payment service during busy sales and restard failed container automatically...